DDI-Net 3.0

Explainable drug interaction intelligence

# Drug–Drug Interaction Research Report

Warfarin + Fluconazole

DRUGS ANALYSED

A **Warfarin**
DB00682 · approved

B **Fluconazole**
DB00196 · approved, investigational

| | |
|---|---|
| Report ID | DDI-20261016-052230-55786C2D |
| Generated | 16 October 2026, 05:22:30 (UTC, UTC+00:00) |
| Generated (UTC) | 2026-10-16T05:22:30.638Z |
| Model | BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0 |
| API report version | 3.0 |
| Analysis produced by the service | 2026-10-16T05:22:29+00:00 |
| Response received by the browser | 2026-10-16T05:22:29.661Z |
| Service | http://127.0.0.1:8765 |
| Language | English |

CONTENTS


1 Executive summary
2 Drug information
3 Documented interactions
4 Biological mechanisms
5 BIO-GINE research analysis
6 Structured explanation from the service
7 Sources and reproducibility
8 Limitations
9 Glossary
S1 Evidence records as returned by the service
A Machine-readable summary (JSON)


RESEARCH USE ONLY

DDI-Net is a research prototype — not a medical device and not clinical decision support. The absence of a documented interaction is not evidence of safety. The BIO-GINE score is a research signal of resemblance to documented pairs, not a probability of clinical harm.

# 1 Executive summary

Documented in a database

## Interaction documented in a database or reference set; no severity grade is shown in this report

A database describes an interaction; it gives no severity.

This is the kind of documentation found, not a level of danger.

INTERACTION STATUS (OUTCOME ENGINE)

## Documented interaction found

Sources name specific consequences or changes in effect for this pair. What they are, and on what basis, is below.

### What could happen?

- **Bleeding** — clinical outcome; documented; severity not established. [2]
- **Higher serum concentration** — change in exposure; documented, mechanism supported by annotations; severity not established. [1]

Seriousness: Severity not established from available evidence.

IN BRIEF

Documentation found — FDA label documents: 0 · distinct excerpts: 0 · DrugBank records: 1 · CRESCENDDI records: 1. Mechanism hypotheses: 3. BIO-GINE calibrated score: 0.81 — a research signal, not a risk.

FOUR KINDS OF INFORMATION IN THIS REPORT

**Documented evidence**

FDA labels: 0 (0 excerpts); DrugBank records: 1

A label or database record that names both drugs.

**Mechanism hypotheses**

hypotheses: 3; shared Reactome pathways: 10

Assembled from each drug's documented roles separately; not tested for the pair.

**Machine-learning prediction**

calibrated score 0.81 · regime S2

BIO-GINE's resemblance to documented pairs; not a probability of harm.

**Missing or unavailable evidence**

FDA labels (openFDA): unavailable

What could not be checked, or is absent from the sources.

## What the official sources document

- DrugBank (TDC export) holds 1 interaction record(s) for this pair. A record gives the interaction type only — no severity, and no note of what it is based on. [1]
- CRESCENDDI reference set: 1 pair–adverse-event record(s). [2]

## What the quoted evidence directly says

- In the sources' own words (shortened where marked "…"; full text in section 3). Only what these sentences state is documented; the report adds no interpretation to them.
- DrugBank, standard sentence; direction not verified
  *"The serum concentration of one of the two drugs can be increased when it is combined with the other drug."* [1]
- CRESCENDDI reference-set record (event as a MedDRA term)
  *"CRESCENDDI lists "Haemorrhage" (MedDRA 10055798) as an adverse event of the interaction between fluconazole and warfarin, mapped from the Micromedex interaction description."* [2]

- Direction: the drug order in the DrugBank export is not verified for this record type, so the report does not say which drug affects which.

### What is only a hypothesis

- Fluconazole → Warfarin: inhibits a protein that handles the other drug (CYP2C19, CYP2C9, CYP3A4). H1 H2 H3
- Assembled from what databases record about each drug separately; not tested for this pair. The arrow is the direction of the hypothesis, not a documented finding.
- A hypothesis agreeing with a DrugBank record is not independent confirmation: the record may rest on the same protein data.
- Shared Reactome pathways: 10 (background, not a cause of interaction).

### What BIO-GINE adds

- Calibrated score 0.81 (regime S2): the pair strongly resembles pairs documented as interacting. The model was not trained on this pair's record. Since documentation already exists, the score adds no new evidence that the drugs interact; it shows whether the model, on its own, would rank the pair with documented ones. M
- Measured quality on the pooled held-out test (S2 and S3 pairs together; no S2-only estimate exists): AUPRC 0.812 ± 0.010. That is an average over the whole test set, not the reliability of this one prediction.
- One drug of this pair was in the validation set, whose pairs were used to choose the model's settings and to fit the calibration, so the test figures describe this pair only approximately.
- The score says nothing about mechanism, direction, severity or what to do, and it is not a probability of harm (more in section 5).

### What remains uncertain

- Source “FDA labels (openFDA)”: unavailable. Its data is not part of this report.
- For 1 of 1 DrugBank records the drug order in the export is not verified, so the report does not say which drug affects which.
- Databases do not record how strongly a drug inhibits or induces a protein, or how much of a drug's handling that protein does.
- The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Section 5 gives the quality measured on the held-out test instead — an average over many pairs, not confidence in this pair.
- No source gives a severity, and none is assigned here.
- Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account.

Terms such as S2, AUPRC or CYP450 are explained in the glossary (section 9).

## 2 Drug information

| FIELD | A · WARFARIN | B · FLUCONAZOLE |
|---|---|---|
| International name (INN) | warfarin | fluconazole |
| Russian name | варфарин (transliteration) | флуконазол (transliteration) |
| DrugBank ID | DB00682 | DB00196 |
| DrugBank groups | approved | approved, investigational |
| Type | small molecule | small molecule |
| Query as entered | DB00682 | DB00196 |
| Matched on | — | — |
| Protein annotations (targets / enzymes / transporters / carriers) | 2 / 6 / 0 / 2 | 1 / 4 / 1 / 0 |
| Model split (seed 0) | val | train |
| Protein data available to the model | yes | yes |
| DrugBank page | https://go.drugbank.com/drugs/DB00682 | https://go.drugbank.com/drugs/DB00196 |

Identifiers: DrugBank. INN: DrugCentral, joined on InChIKey (no value means no match). Russian names: rule-based transliteration — an interface label, not sourced data.

# 3 Documented interactions

Evidence is grouped by source document: [n] is a document, [n.k] one distinct excerpt of it; the explanation refers to these labels. Several excerpts of one label are one source, not separate confirmations. The English text is the source's own words (an excerpt or DrugBank's standard sentence).

**FDA label documents: 0 · distinct excerpts: 0 · DrugBank records: 1 · CRESCENDDI records: 1**

## Source documents

| # | DOCUMENT | IDENTIFIER | VERSION / DATE | EXCERPTS | SECTIONS |
|---|---|---|---|---|---|
| [1] | DrugBank (TDC export) | DB00682\|DB00196\|Y=73 | PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank') | 1 | — |
| [2] | CRESCENDDI (reference set; BNF, Micromedex, ANSM, 2018–2019) | positive_control:DB00196\|DB00682\|MedDRA 10055798 | elpidakon/CRESCENDDI@1734d7c12ac1 (data records, index crescenddi_index/v1) | 1 | — |

### [1] DrugBank (TDC export)

Retrieved 2026-08-24

1 interaction record(s); DrugBank records are never merged

DrugBank DDI records via Therapeutics Data Commons · PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank')

Database · DrugBank | direction not verified | Source's words · DrugBank template, direction-neutral

*“The serum concentration of one of the two drugs can be increased when it is combined with the other drug.”*

direction not verified: the report does not say which drug affects which. Order check for this type: 1,270 of 12,308 checkable records consistent (10.3%).

DOCUMENT PROVENANCE

| | |
|---|---|
| URL | https://tdcommons.ai/multi_pred_tasks/ddi/ |
| Record ID | DB00682\|DB00196\|Y=73 |
| Retrieved | 2026-08-24 |
| Licence | TDC: CC BY 4.0; underlying DrugBank content: DrugBank non-commercial licence |
| Severity as stated by the source | not stated by the source |

QUALIFIERS

- DrugBank's standard sentence for this interaction type, not a quotation from a study or a label.
- This export carries no severity grade. None is implied.
- The export does not state the basis of each record (clinical study, label text, or inference from pharmacological properties).
- Drug order for this interaction type matched documented DrugBank roles in only 1,270 of 12,308 checkable records, so the statement is shown without naming which drug is affected.

### [2] CRESCENDDI (reference set; BNF, Micromedex, ANSM, 2018–2019)

Retrieved 2026-10-16

1 pair–adverse-event record(s)

CRESCENDDI reference set (Kontsioti et al., Sci Data 2022) · elpidakon/CRESCENDDI@1734d7c12ac1 (data records, index crescenddi_index/v1)

Reference set · CRESCENDDI | Source's words · DrugBank template, names filled in

*“CRESCENDDI lists “Haemorrhage” (MedDRA 10055798) as an adverse event of the interaction between fluconazole and warfarin, mapped from the Micromedex interaction description.”*

DOCUMENT PROVENANCE

| | |
|---|---|
| URL | https://github.com/elpidakon/CRESCENDDI |
| Record ID | positive_control:DB00196\|DB00682\|MedDRA 10055798 |
| Retrieved | 2026-10-16 |
| Licence | MIT License (authors' GitHub repository, which contains the data records); figshare licence not verified from the build environment |
| Severity as stated by the source | graded by the compendia, but those grades are not reproduced here |

QUALIFIERS

- CRESCENDDI positive control: the pair is documented as interacting in all three compendia the authors used (BNF, Nov 2018; IBM Micromedex, Sep 2018; ANSM Thesaurus, Sep 2019).
- The adverse event is the MedDRA term the authors mapped from the compendium's interaction description; the compendium text itself is not reproduced.
- The compendia grade severity, but those grades are proprietary editorial classifications and are not shown here. Severity is therefore not established from the evidence shown.
- Snapshot of 2018–2019 sources; later changes to the compendia are not reflected.

## Clinical outcomes (as the sources state them)

Consequences the sources state for this pair, each with its quotation, document, section and extraction rule. Seriousness only as a source states it.

| OUTCOME | KIND / STATUS | SERIOUSNESS | SOURCES |
|---|---|---|---|
| Bleeding | clinical outcome<br>documented | severity not established | [2] |
| Higher serum concentration | change in exposure<br>documented, mechanism supported by annotations | severity not established | [1] |

clinical outcome · documented · severity not established

### Bleeding

The reference set names this consequence for this pair. It gives no frequency; the severity grades the compendia give are not reproduced here.
Severity not established from available evidence. The compendia behind CRESCENDDI grade severity, but their grades are not reproduced here. The source does not say which drug is affected (or its order is unverified); the report does not name one.
CRESCENDDI reference set: the pair is documented in three clinical compendia (2018–2019) and the event comes from their interaction description. The compendium text is not reproduced. Source documents: 1.
`outcome:DB00196+DB00682:bleeding_coagulation:bleeding`

1. CRESCENDDI reference set (Kontsioti et al., Sci Data 2022) · 2026-10-16

*“CRESCENDDI lists “Haemorrhage” (MedDRA 10055798) as an adverse event of the interaction between fluconazole and warfarin, mapped from the Micromedex interaction description.”*

`meddra_effect_map/v1:meddra=10055798 · Haemorrhage · positive_control:DB00196|DB00682|MedDRA 10055798`
CRESCENDDI positive control: the pair is documented as interacting in all three compendia the authors used (BNF, Nov 2018; IBM Micromedex, Sep 2018; ANSM Thesaurus, Sep 2019).
The adverse event is the MedDRA term the authors mapped from the compendium's interaction description; the compendium text itself is not reproduced.
The compendia grade severity, but those grades are proprietary editorial classifications and are not shown here. Severity is therefore not established from the evidence shown.
Snapshot of 2018–2019 sources; later changes to the compendia are not reflected.

change in exposure · documented, mechanism supported by annotations · severity not established

### Higher serum concentration · Warfarin

The source describes a change in how much of a drug is in the body. That, by itself, is not a documented clinical outcome.
Severity not established from available evidence. By the protein roles, the affected drug is Warfarin. This comes from the annotations (inhibitor/inducer and substrate), not from the DrugBank record's order.
DrugBank standard sentence; the source does not state what the record is based on. Source documents: 1.
`outcome:DB00196+DB00682:exposure_increase:serum_concentration_increase`

1. DrugBank DDI records via Therapeutics Data Commons · 2026-08-24

*“The serum concentration of #Drug2 can be increased when it is combined with #Drug1.”*

drugbank_type_map/v1:type=73 · serum concentration · DB00682|DB00196|Y=73
DrugBank gives no severity and does not state the basis of the record.

• A change in concentration is not, by itself, a documented clinical outcome.

### Sources checked

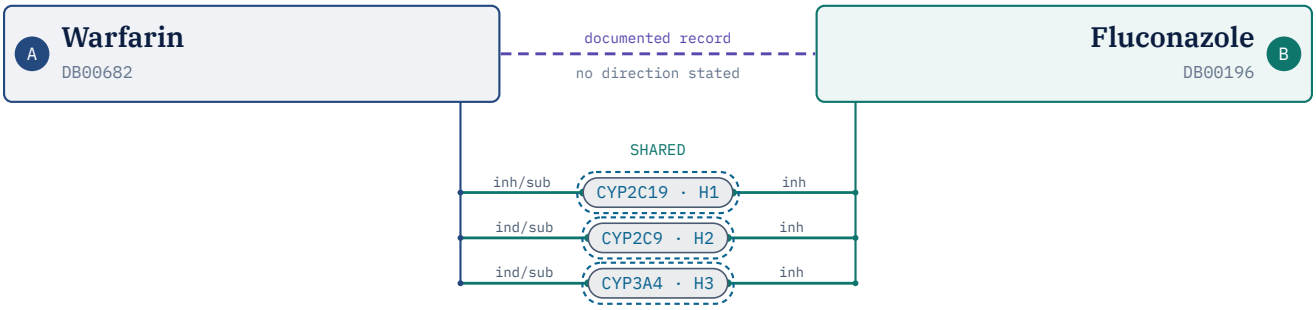
| SOURCE | STATUS | DETAIL |
|---|---|---|
| DrugBank (TDC export, local) | checked | 1 record(s) in the local DrugBank export (TDC, retrieved 2026-08-24). |
| CRESCENDDI (reference set, 2018–2019) | checked | 1 pair-event record(s) in the CRESCENDDI reference set (positive controls, compendia of 2018–2019). |
| FDA labels (openFDA) | unavailable | openfda_label: network error (URLError) |

## 4 Biological mechanisms

Below are each drug's documented roles on their own (DrugBank, ChEMBL) and how they line up. These are mechanism hypotheses, not evidence of an interaction: the databases do not record strength of effect or how much of a drug's processing a protein handles.

### Mechanism graph

Drawn from the report's mechanism block: proteins documented for both drugs. The full graph, including proteins of only one drug, is on the Mechanisms page.




Roles on branches: sub — substrate, inh — inhibitor / inhibits, ind — inducer, act — activates, n/s — not stated.

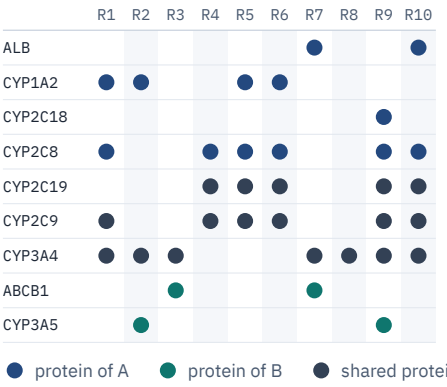
The arc between the drugs is a documented interaction record. It has no arrowhead: the DrugBank export's drug order is not verified for this type.

### Mechanism hypotheses

| # | PROTEIN | CONFIGURATION | ROLES A / B | BASIS |
|---|---|---|---|---|
| H1 | CYP2C19<br>Cytochrome P450 2C19<br>P33261 · enzyme | inhibits a protein that handles the other drug<br>hypothesis: Fluconazole → Warfarin<br>consistent with record [1] (not independent confirmation) | A: inhibitor, substrate<br>B: inhibitor | DrugBank drug-protein annotation 5.1 (export 2025-01-02)<br>PMID 16141567, 10709776, 19515014, 16162970, 16372822 |
| H2 | CYP2C9<br>Cytochrome P450 2C9<br>P11712 · enzyme | inhibits a protein that handles the other drug<br>hypothesis: Fluconazole → Warfarin<br>consistent with record [1] (not independent confirmation) | A: inducer, substrate<br>B: inhibitor | DrugBank drug-protein annotation 5.1 (export 2025-01-02)<br>PMID 16141567, 8801056, 10952477, 11475469, 10709776, 29283396, 16372822, 20735727 |
| H3 | CYP3A4<br>Cytochrome P450 3A4<br>P08684 · enzyme | inhibits a protein that handles the other drug<br>hypothesis: Fluconazole → Warfarin<br>consistent with record [1] (not independent confirmation) | A: inducer, substrate<br>B: inhibitor | DrugBank drug-protein annotation 5.1 (export 2025-01-02)<br>PMID 16141567, 8801056, 15969931, 19515014, 16372822, 20735727 |

## Shared biological pathways (Reactome) · 10

10 of 10 shown: pathways through a shared protein first, then the most specific. Protein membership in Reactome pathways (a dot means the protein is in the pathway; colour shows whose protein it is: A, B or shared).




| R | PATHWAY | REACTOME | PROTEINS | VIA A | VIA B |
|---|---|---|---|---|---|
| R1 | Biosynthesis of maresin-like SPMs | https://reactome.org/content/detail/R-HSA-9027307 | 6 | CYP1A2, CYP2C8, CYP2C9, CYP3A4 | CYP2C9, CYP3A4 |
| R2 | Aflatoxin activation and detoxification | https://reactome.org/content/detail/R-HSA-5423646 | 8 | CYP1A2, CYP3A4 | CYP3A4, CYP3A5 |
| R3 | Atorvastatin ADME | https://reactome.org/content/detail/R-HSA-9754706 | 8 | CYP3A4 | ABCB1, CYP3A4 |
| R4 | CYP2E1 reactions | https://reactome.org/content/detail/R-HSA-211999 | 8 | CYP2C19, CYP2C8, CYP2C9 | CYP2C19, CYP2C9 |
| R5 | Synthesis of (16-20)-hydroxyeicos-atetraenoic acids (HETE) | https://reactome.org/content/detail/R-HSA-2142816 | 8 | CYP1A2, CYP2C19, CYP2C8, CYP2C9 | CYP2C19, CYP2C9 |
| R6 | Synthesis of epoxy (EET) and dihydroxyeicosatrienoic acids (DHET) | https://reactome.org/content/detail/R-HSA-2142670 | 8 | CYP1A2, CYP2C19, CYP2C8, CYP2C9 | CYP2C19, CYP2C9 |
| R7 | Prednisone ADME | https://reactome.org/content/detail/R-HSA-9757110 | 10 | ALB, CYP3A4 | ABCB1, CYP3A4 |
| R8 | Phase I - Functionalization of compounds | https://reactome.org/content/detail/R-HSA-211945 | 13 | CYP3A4 | CYP3A4 |
| R9 | Xenobiotics | https://reactome.org/content/detail/R-HSA-211981 | 16 | CYP2C18, CYP2C19, CYP2C8, CYP2C9, CYP3A4 | CYP2C19, CYP2C9, CYP3A4, CYP3A5 |
| R10 | Aspirin ADME | https://reactome.org/content/detail/R-HSA-9749641 | 25 | ALB, CYP2C19, CYP2C8, CYP2C9, CYP3A4 | CYP2C19, CYP2C9, CYP3A4 |

### From mechanism to outcome

Each link is marked: directly documented, verified annotation, or plausible but unconfirmed. A shared protein or pathway alone is never a link.

directly documented — verified biological annotation - - - plausible but unconfirmed ····· unsupported or unavailable




Fluconazole inhibits CYP2C19, which processes Warfarin (annotations). This drug's exposure change is documented for the pair.
verified biological annotation: Fluconazole is annotated as an inhibitor of CYP2C19.
verified biological annotation: Warfarin is annotated as a substrate of CYP2C19.
directly documented: A change in this drug's exposure is documented for the pair.
plausible but unconfirmed: The outcome is documented for the pair, but that it arises through this route is not confirmed.




Fluconazole inhibits CYP2C9, which processes Warfarin (annotations). This drug's exposure change is documented for the pair.
verified biological annotation: Fluconazole is annotated as an inhibitor of CYP2C9.
verified biological annotation: Warfarin is annotated as a substrate of CYP2C9.
directly documented: A change in this drug's exposure is documented for the pair.
plausible but unconfirmed: The outcome is documented for the pair, but that it arises through this route is not confirmed.




Fluconazole inhibits CYP3A4, which processes Warfarin (annotations). This drug's exposure change is documented for the pair.
verified biological annotation: Fluconazole is annotated as an inhibitor of CYP3A4.
verified biological annotation: Warfarin is annotated as a substrate of CYP3A4.
directly documented: A change in this drug's exposure is documented for the pair.
plausible but unconfirmed: The outcome is documented for the pair, but that it arises through this route is not confirmed.

### Proteins in the hypotheses: family background (project text, not a statement about this pair)

| | |
|---|---|
| CYP2C19 | An enzyme of the cytochrome P450 family. These work mainly in the liver and gut and break down many medicines so the body can clear them. |
| CYP2C9 | An enzyme of the cytochrome P450 family. These work mainly in the liver and gut and break down many medicines so the body can clear them. |
| CYP3A4 | An enzyme of the cytochrome P450 family. These work mainly in the liver and gut and break down many medicines so the body can clear them. |

## 5 BIO-GINE research analysis

| CALIBRATED SCORE | RAW SCORE | REGIME | RELIABILITY FOR THIS PAIR |
|---|---|---|---|
| 0.81 | > 0.9999 | S2 | like the held-out pairs of the frozen test |

Numbers are rounded; exact values are in Appendix A and the companion JSON.

BIO-GINE is a research model that looks for patterns resembling documented interactions. Its score is not a probability of harm or an individual's risk, and it does not affect the status or the outcomes above.

| | |
|---|---|
| Model and checkpoint | BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0 |
| Regime | S2 — one drug was outside the training set |
| Splits A / B | val / train |
| Pair's record seen in training | no |
| Protein data available to the model | A: yes · B: yes |

**What the two numbers mean**

- Raw score (> 0.9999): the network's direct output, a number between 0 and 1. Near the ends of the scale the values bunch up: differences in the fourth or fifth decimal place mean nothing beyond “very high” (or “very low”).

- Calibrated score (0.81): the same output, rescaled by “temperature scaling”. The rescaling was fitted on validation pairs of which half were documented interactions and half random pairs. It pulls over-confident values back from the edges of the scale but does not change the order of pairs. That is why the two numbers differ.
- Neither number is the probability that the combination harms anyone. The model only learned to tell documented pairs from random ones. The calibrated score could be read as a share of documented pairs only in a half-and-half mix like the one it was fitted on; real drug combinations are not such a mix, and a documented interaction says nothing about how serious it is.
- What they can tell: how strongly the model groups this pair with pairs already documented as interacting. That is useful for ranking pairs for research.
- What they cannot tell: whether an interaction happens in a particular person, how strong or serious it would be, which drug affects which, or what anyone should do.

## Measured model quality (held-out test set, not this pair)

| METRIC | MEAN ± SD | SEEDS | TEST SET | SOURCE |
|---|---|---|---|---|
| AUPRC | 0.812 ± 0.010 | 5 | pooled drug-disjoint test (S2 and S3; 91% S2) | paper/FACTS.json models.M4_pooled (v2_final_test.csv @ frozen tag) |
| AUROC | 0.818 ± 0.010 | 5 | pooled, both annotated | reports/v3_coverage_reliability.json |

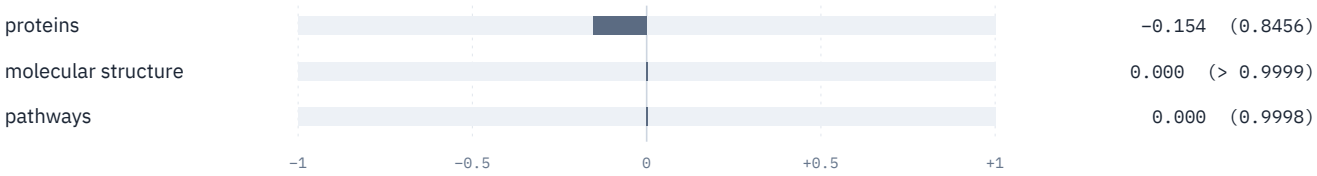
- This pair's regime is S2. The frozen results evaluate S2 pairs only together with S3 pairs: the pooled test has 84,690 pairs, 76,932 of them S2 and 7,758 S3. No S2-only estimate exists, so the pooled figure is shown and labelled as pooled.
- “both annotated”: only test pairs whose two drugs both have protein data (79,712 pairs of the pooled test).
- AUPRC and AUROC describe the model's quality over thousands of test pairs; “±” is the spread between 5 training runs. They do not say how likely this particular prediction is to be right. The service gives no uncertainty for a single pair: it uses one trained copy of the model (seed 0).
- One drug of this pair was in the validation set, whose pairs were used to choose the model's settings and to fit the calibration, so the test figures describe this pair only approximately.

## Uncertainty

The service uses one model checkpoint (seed 0) and provides neither a per-pair confidence interval nor an ensemble estimate. Section 5 gives the quality measured on the held-out test instead — an average over many pairs, not confidence in this pair.

## What the model leans on (one kind of data zeroed)

Change in the raw score when one kind of data is removed for both drugs. A minus sign means the score drops without that data. In brackets: the raw score without this data.




Raw score: > 0.9999

The raw score is near the edge of the scale (0 or 1); changes are compressed there, and reliance can look smaller than it is.

This shows which data the model leans on for this pair, not why the drugs interact: the model never saw zeroed inputs in training.

Method check: reproduces the frozen V2 analysis on 84,690 test rows, max |Δ| = 6.1e-6.

Method: branch zeroed before the frozen fusion layer for both drugs; mask unchanged (as scripts/49_v2_interpretability.py at the frozen tag)

## Notes from the service

- One drug was in the validation set: the model was not trained on it, but it was used to choose the configuration and the calibration.
- The calibrated score was fitted on validation pairs with equal numbers of documented and randomly drawn pairs. It is not a probability of harm, nor how often interactions happen in practice.

# 6 Structured explanation from the service

The text below was written by the service with deterministic templates over the retrieved records (no language model) and checked by it for advice wording. Labels point to records: [n] source document, [n.k] one of its excerpts, H hypothesis, M model, R Reactome pathway. The text below comes from the service; its terms (S2, AUPRC, AUROC) are explained in the glossary (section 9).

## What is known

Source — DrugBank describes this combination as: “The serum concentration of one of the two drugs can be increased when it is combined with the other drug.” [1]

Source — The CRESCENDDI reference set (built from the BNF, Micromedex and ANSM compendia, 2018–2019) names 1 adverse event(s) for this pair: “Haemorrhage” (MedDRA terms). [2]

Caveat — For this record type the drug order in the DrugBank export proved unreliable (we measured it), so the text does not say which drug affects which. [1]

## Why this may happen

Hypothesis — fluconazole is recorded as inhibiting CYP2C19, and warfarin as something CYP2C19 processes. With CYP2C19 inhibited, warfarin may be broken down more slowly than usual. H1

Hypothesis — fluconazole is recorded as inhibiting CYP2C9, and warfarin as something CYP2C9 processes. With CYP2C9 inhibited, warfarin may be broken down more slowly than usual. H2

Hypothesis — fluconazole is recorded as inhibiting CYP3A4, and warfarin as something CYP3A4 processes. With CYP3A4 inhibited, warfarin may be broken down more slowly than usual. H3

Caveat — These hypotheses agree with a DrugBank interaction record. That is not independent confirmation: DrugBank may have derived the record from the same protein annotations. [1] H1 H2 H3

Caveat — The databases record neither how strongly a drug inhibits or induces a protein nor how much of a drug's handling that protein does. These are hypotheses, not a conclusion that an interaction will happen. H1 H2 H3

Annotation — Proteins of both drugs take part in 10 shared biological process(es) in Reactome — for example “Biosynthesis of maresin-like SPMs”. A shared process is background, not a cause of interaction. R

## What the AI found

Model — BIO-GINE finds this pair strongly resembles pairs already recorded as interacting. This is resemblance to recorded pairs, not an estimate of harm. M

Model — For this pair the model's score depends most on the protein data: without it the score moves by 0.15. This says what the model leans on, not why the drugs interact. M

Caveat — One drug was in the validation set: the model was not trained on it, but it was used to choose the configuration and the calibration. M

Caveat — The calibrated score was fitted on validation pairs with equal numbers of documented and randomly drawn pairs. It is not a probability of harm, nor how often interactions happen in practice. M

## How certain is this?

Source — Documented: 1 DrugBank record(s) and 1 CRESCENDDI reference-set record(s). Several excerpts of one label are one source, not separate confirmations, and a DrugBank record may itself rest on label text. This is the most reliable part of the page. [1] [2]

Caveat — No severity is assigned: the DrugBank export has none, and for labels only the section that names the drug is shown.

Caveat — The compendia behind CRESCENDDI grade interaction severity, but those grades are their own editorial classifications and are not reproduced here.

Hypothesis — The 3 mechanism hypothesis(es) are assembled from each drug's documented roles separately. None was tested for this pair. H1 H2 H3

Model — Model quality on the pooled held-out test (S2 and S3 pairs together, 91% S2; the frozen results hold no S2-only figure): AUPRC 0.812 ± 0.010 (spread across 5 training runs). That is an average over many pairs, not confidence in this particular pair. M

### What is missing

Missing — The source “FDA labels (openFDA)” could not be checked right now. Its data is not part of this answer.

Caveat — Doses, duration, age, kidney and liver function, other medicines and individual factors are not taken into account — everything that decides whether an interaction shows up and how strongly.

Caveat — Decisions about medicines belong to a doctor or pharmacist. This page cannot replace them.

Method: deterministic templates over retrieved records; no language model

## 7 Sources and reproducibility

### Evidence records

| # | SOURCE | IDENTIFIER | RETRIEVED | URL |
|---|---|---|---|---|
| [1] | DrugBank DDI records via Therapeutics Data Commons · PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank') | DB00682\|DB00196\|Y=73 | 2026-08-24 | https://tdcommons.ai/multi_pred_tasks/ddi/ |
| [2] | CRESCENDDI reference set (Kontsioti et al., Sci Data 2022) · elpidakon/CRESCENDDI@1734d7c12ac1 (data records, index crescenddi_index/v1) | positive_control: DB00196\|DB00682\|MedDRA 10055798 | 2026-10-16 | https://github.com/elpidakon/CRESCENDDI |

### Protein annotations the hypotheses rest on

| DRUG | PROTEIN | ROLES | SOURCE | RETRIEVED | PUBMED |
|---|---|---|---|---|---|
| Fluconazole | CYP2C19 P33261 | inhibitor raw action: inhibitor | DrugBank drug-protein annotation · 5.1 (export 2025-01-02) | 2026-08-28 | 16141567, 10709776 |
| Warfarin | CYP2C19 P33261 | inhibitor, substrate raw action: substrate\|inhibitor | DrugBank drug-protein annotation · 5.1 (export 2025-01-02) | 2026-08-28 | 19515014, 16162970, 16372822 |
| Fluconazole | CYP2C9 P11712 | inhibitor raw action: inhibitor | DrugBank drug-protein annotation · 5.1 (export 2025-01-02) | 2026-08-28 | 16141567, 8801056, 10952477, 11475469, 10709776 |
| Warfarin | CYP2C9 P11712 | inducer, substrate raw action: substrate\|inducer | DrugBank drug-protein annotation · 5.1 (export 2025-01-02) | 2026-08-28 | 29283396, 16372822, 20735727 |
| Fluconazole | CYP3A4 P08684 | inhibitor raw action: inhibitor | DrugBank drug-protein annotation · 5.1 (export 2025-01-02) | 2026-08-28 | 16141567, 8801056, 15969931 |
| Warfarin | CYP3A4 P08684 | inducer, substrate raw action: substrate\|inducer | DrugBank drug-protein annotation · 5.1 (export 2025-01-02) | 2026-08-28 | 19515014, 16372822, 20735727 |

PubMed: https://pubmed.ncbi.nlm.nih.gov/<PMID>/ · UniProt: https://www.uniprot.org/uniprotkb/<accession>/entry

### Provenance and parameters

| | |
|---|---|
| Endpoint (report) | http://127.0.0.1:8765/api/v3/report |
| Request body (report) | {"drug_a":"DB00682","drug_b":"DB00196","include_labels":true} |
| Response received by the browser (report) | 2026-10-16T05:22:29.661Z |

| | |
|---|---|
| Report generated by the service (generated_at) | 2026-10-16T05:22:29+00:00 |
| API report version | 3.0 |
| Explanation method | deterministic templates over retrieved records; no language model |
| Model | BIO-GINE M4 · bd45f84e3c1b2c33 · seed 0 |
| Frozen research tag | `v2-final-github-safe-2026-09-03 · 92c481eeaba8faff991ced850e1c4de418ea31b0` |
| Site | http://127.0.0.1:4173 |
| Site build commit | not recorded |
| PDF generator | DDI-Net web, in-browser export (@react-pdf/renderer) |
| Generated | 16 October 2026, 05:22:30 (UTC, UTC+00:00) |
| Generated (UTC) | 2026-10-16T05:22:30.638Z |
| Language | English |
| SHA-256 of the response data | `55786c2d08a3f26d5bf589defcad1452371f08aa7526224a299d0de22ba68b91`<br>Hash of the service response(s) as canonical JSON (keys sorted, serialised with JSON.stringify). The same hash is in the companion JSON file, available from the “JSON” button next to “Export PDF”. |

Retrieval times are given per record (retrieved_at) as returned by the service: the export date for local extracts, the request date for labels.

# 8 Limitations

## Core limitations

- DDI-Net is a research tool, not clinical decision support and not a medical device. Decisions about medicines belong to a doctor or pharmacist.
- Unlabelled pairs are not confirmed safe. A missing record means only that the sources checked do not contain one.
- BIO-GINE scores are not calibrated probabilities of clinical harm: temperature scaling was fitted on validation pairs at 50% prevalence against randomly drawn unlabelled pairs.
- The drug order in the DrugBank (TDC) export was verified by measurement for only two interaction types; for the others it is either meaningless (symmetric types) or unverified, so directional claims (“A raises the level of B”) are not made.
- External medical evidence (FDA labels, RxNorm) may be incomplete or unavailable at query time; its status is listed under “Sources checked”.

## Limitations reported by the service for this report

- Only the 1,705 drugs of the research dataset; 1,497 of them have an international name.
- DrugBank records carry no severity and do not state their basis. For many record types the drug order in the export is unreliable (measured).
- Labels are searched by drug name. Statements about whole drug classes (“strong CYP3A4 inhibitors”) are not recognised, so no match does not mean the label is silent.
- Mechanism hypotheses are assembled from each drug's roles separately; strength of effect and the protein's share of the work are unknown.
- The BIO-GINE score is a research signal of resemblance to documented pairs. It is not a probability of harm.

> RESEARCH USE ONLY
> This is a research prototype, not a medical device. It does not replace a doctor or pharmacist and cannot say whether a combination is safe for a particular person.

# 9 Glossary

Plain explanations of the terms used in this report.

| | |
|---|---|
| **Source evidence vs model prediction** | Source evidence is text that an official label or a database actually contains about these two drugs. A model prediction is a number BIO-GINE computes from patterns in the data it was trained on. The first records what is known; the second is a research guess. |
| **FDA label (SPL)** | A medicine's official prescribing information in the United States, published by the FDA as Structured Product Labeling. One label is one document with its own set ID, divided into sections such as Warnings or Drug Interactions. |
| **Document and excerpt** | An excerpt is one sentence quoted from a document. Several excerpts of one label are still one source and do not confirm each other. When a label repeats a sentence in two sections, it is shown once. |
| **DrugBank (TDC export)** | A large database of drugs and known interactions; DDI-Net uses the public copy distributed by Therapeutics Data Commons. A record states only an interaction type, in a standard sentence, with no severity and no note of what it is based on. |
| **Direction** | Which drug affects which. For most interaction types the drug order in the DrugBank export is unreliable, so the report names no direction from DrugBank. Label sentences are quoted as written; no direction is inferred from the mere fact that one drug's label names the other. |
| **Cytochrome P450 (CYP450)** | A family of enzymes, mostly in the liver (for example CYP2C9, CYP3A4), that break down many medicines. If one drug slows an enzyme that breaks down another, the second may stay in the body longer. |
| **Substrate, inhibitor, inducer** | A substrate is a drug a protein processes. An inhibitor slows the protein down. An inducer makes the body produce more of it. |
| **Mechanism hypothesis** | A possible biological explanation assembled from what databases record about each drug separately (for example: drug X inhibits an enzyme that processes drug Y). It was not tested for this pair and says nothing about the size of any effect. |
| **Reactome pathway** | A curated chain of reactions in the cell. Two drugs sharing a pathway is background information, not a cause of interaction. |
| **Raw and calibrated score** | The raw score is the model's direct output between 0 and 1; the calibrated score is the same output rescaled on balanced validation pairs. Both put pairs in the same order. Neither is a probability of harm. |
| **Regimes S1, S2, S3** | Whether the model saw the drugs during training. S1: both drugs were in training (quality not measured). S2: one drug was not in training. S3: neither was — the hardest test. |
| **Held-out test; “pooled”** | Pairs set aside in advance and never used for training; quality is measured on them. “Pooled” means S2 and S3 test pairs measured together (84,690 pairs, 7,758 of them S3). |
| **AUROC** | The chance that the model ranks a randomly chosen documented pair above a randomly chosen pair from the comparison set (random pairs without a record). 0.5 is guessing; 1 is a perfect ranking. |
| **AUPRC** | Area under the precision–recall curve: how well the model finds documented pairs without raising many false alarms, over all thresholds at once. Half of the test pairs are documented, so guessing gives about 0.5; 1 is perfect. |
| **± (spread between runs)** | The model was trained 5 times, each from a different random starting point (seed). “±” is the standard deviation of the result between those runs. It is not an error bar for a single pair. |

## S1 Supplement S1. Evidence records as returned by the service

Every record of the API response, with its section, identifiers and retrieval time. Records that repeat one label's text map to the same excerpt; the full response is in the companion JSON file.

**[1]** `drugbank-tdc:DB00682>DB00196:t73`

| | |
|---|---|
| Category | Database · DrugBank |
| Source | DrugBank DDI records via Therapeutics Data Commons · PyTDC 1.1.15, tdc.multi_pred.DDI(name='DrugBank') |
| Record ID | `DB00682\|DB00196\|Y=73` |
| Original identifiers | `drugbank_id_drug1_as_exported: DB00682`<br>`drugbank_id_drug2_as_exported: DB00196`<br>`tdc_interaction_type: 73` |
| URL | https://tdcommons.ai/multi_pred_tasks/ddi/ |
| Retrieved | 2026-08-24 |
| DrugBank template | The serum concentration of #Drug2 can be increased when it is combined with #Drug1. |
| Direction | direction not verified: the report does not say which drug affects which |
| Licence | TDC: CC BY 4.0; underlying DrugBank content: DrugBank non-commercial licence |

**[2]** `crescenddi:DB00196+DB00682:10055798`

| | |
|---|---|
| Category | Reference set · CRESCENDDI |
| Source | CRESCENDDI reference set (Kontsioti et al., Sci Data 2022) · elpidakon/CRESCENDDI@1734d7c12ac1 (data records, index crescenddi_index/v1) |
| Record ID | `positive_control:DB00196\|DB00682\|MedDRA 10055798` |
| Original identifiers | `meddra_code: 10055798`<br>`meddra_term: Haemorrhage`<br>`event_source: Micromedex`<br>`effect_key: bleeding`<br>`event_map: meddra_effect_map/v1` |
| URL | https://github.com/elpidakon/CRESCENDDI |
| Retrieved | 2026-10-16 |
| DrugBank template | CRESCENDDI positive control: drug pair + MedDRA event + event source |
| Direction | symmetric type: direction carries no meaning |
| Licence | MIT License (authors' GitHub repository, which contains the data records); figshare licence not verified from the build environment |

## A Appendix A. Machine-readable summary (JSON)

Key fields as JSON for automated checking. The full service responses are in the companion JSON file (same SHA-256).


```
{
  "schema": "ddinet.pdf-summary/1",
  "report_id": "DDI-20261016-052230-55786C2D",
  "generated_at": "2026-10-16T05:22:30+00:00",
  "generated_at_utc": "2026-10-16T05:22:30.638Z",
  "time_zone": "UTC",
  "language": "en",
  "payload_sha256": "55786c2d08a3f26d5bf589defcad1452371f08aa7526224a299d0de22ba68b91",
  "inputs": [
    {"endpoint":"http://127.0.0.1:8765/api/v3/report","request":{"drug_a":"DB00682","drug_b":"DB00196","include_labels":true},
```

```
      "received_at":"2026-10-16T05:22:29.661Z"}
  ],
  "server_generated_at": "2026-10-16T05:22:29+00:00",
  "report_version": "3.0",
  "drugs": [
    {"role":"A","id":"DB00682","inn":"warfarin","name_ru":"варфарин"},
    {"role":"B","id":"DB00196","inn":"fluconazole","name_ru":"флуконазол"}
  ],
  "evidence_status": "documented_no_severity",
  "evidence_counts": {
    "labelDocuments": 0,
    "labelExcerpts": 0,
    "noEffectExcerpts": 0,
    "labelRecords": 0,
    "mergedRepeats": 0,
    "databaseRecords": 1,
    "referenceRecords": 1
  },
  "source_documents": [
    {"ref":"[1]","kind":"database","document_id":"drugbank_tdc","owner":null,"mentions":null,"excerpts":[{"ref":"[1]","category":
      "database_interaction_record","sections":[],"records":["drugbank-tdc:DB00682>DB00196:t73"]}]},
    {"ref":"[2]","kind":"reference","document_id":"crescenddi","owner":null,"mentions":null,"excerpts":[{"ref":"[2]","category":
      "reference_set_outcome","sections":[],"records":["crescenddi:DB00196+DB00682:10055798"]}]}
  ],
  "evidence": [
    {"ref":"[1]","id":"drugbank-tdc:DB00682>DB00196:t73","category":"database_interaction_record","source":
      "DrugBank DDI records via Therapeutics Data Commons","record_id":"DB00682|DB00196|Y=73","section":null,"also_in_sections":[],
      "direction":"not_verified","retrieved_at":"2026-08-24","url":"https://tdcommons.ai/multi_pred_tasks/ddi/"},
    {"ref":"[2]","id":"crescenddi:DB00196+DB00682:10055798","category":"reference_set_outcome","source":
      "CRESCENDDI reference set (Kontsioti et al., Sci Data 2022)","record_id":"positive_control:DB00196|DB00682|MedDRA 10055798",
      "section":null,"also_in_sections":[],"direction":"symmetric","retrieved_at":"2026-10-16","url":
      "https://github.com/elpidakon/CRESCENDDI"}
  ],
  "conflicts": 0,
  "hypotheses": [
    {"ref":"H1","gene":"CYP2C19","uniprot":"P33261","relation":"enzyme","pattern":"inhibitor_substrate","actor":"DB00196",
      "affected":"DB00682"},
    {"ref":"H2","gene":"CYP2C9","uniprot":"P11712","relation":"enzyme","pattern":"inhibitor_substrate","actor":"DB00196","affected":
      "DB00682"},
    {"ref":"H3","gene":"CYP3A4","uniprot":"P08684","relation":"enzyme","pattern":"inhibitor_substrate","actor":"DB00196","affected":
      "DB00682"}
  ],
  "shared_pathways_total": 10,
  "model": {
    "model": "BIO-GINE M4",
    "checkpoint": "bd45f84e3c1b2c33",
    "available": true,
    "calibrated_score": 0.8108211344965719,
    "raw_score": 0.9999718803394634,
    "regime": "S2",
    "reliability": "held_out_like",
    "abstain": false
  },
  "sources_checked": [
    {"source":"drugbank_tdc","status":"ok"},
    {"source":"crescenddi","status":"ok"},
    {"source":"openfda_label","status":"unavailable"}
  ],
  "outcomes": {
    "interaction_status": "documented_interaction",
    "seriousness": "documented_no_standard_severity",
    "records": [{"id":"outcome:DB00196+DB00682:bleeding_coagulation:bleeding","category":"bleeding_coagulation","effect":"bleeding",
      "level":"clinical_outcome","status":"documented","seriousness":"documented_no_standard_severity","research_only":false,
      "affected":null,"evidence":["crescenddi:DB00196+DB00682:10055798"],"methods":["meddra_effect_map/v1:meddra=10055798"]},{"id":
      "outcome:DB00196+DB00682:exposure_increase:serum_concentration_increase","category":"exposure_increase","effect":
      "serum_concentration_increase","level":"pharmacokinetic_change","status":"mechanistically_supported","seriousness":
      "documented_no_standard_severity","research_only":false,"affected":"DB00682","evidence":["drugbank-tdc:DB00682>DB00196:t73"],
      "methods":["drugbank_type_map/v1:type=73"]}]
  }
}
```